Ion CSA v3 response study

October 30, 2026

The study processes 8,000 event IDs per sample at 4 mm pitch and 40 ms frame width. Every pixel with charge retains its complete event history, including empty frames, two quiet leading frames and a three-second recovery window. There is no S/N cut. Signed reconstructed counts, including negative noise fluctuations, are retained.

The circuit is the provisional v3 design, nominal 20 s discharge, with 5 ms centred voltage averages, 1.25 pF external capacitance and 3 fF feedback. The input uses uniform mean collection within each frame and the established central 0.5 mm square-pad weighting benchmark in an infinite grounded plane. The pad weighting potential is one, with zero on the surrounding plane and at infinity, in homogeneous dielectric. The ion speed is 3 m/s through the finite 5 mm gap. The charge integral is $e[1 - \phi_w(g)]$ per collected ion.

The same transistor circuit was recalibrated at 20, 30 and 40 ms. Noise uses its one-sided stationary operating-point spectrum from 0.0001 Hz to 1 MHz, filtered by the 5 ms average and the decay-corrected frame difference. Gaussian samples retain temporal covariance; pad channels are independent. This stationary noise approximation is also applied to compressed mean signals.

Frame (ms)	Fitted τ (s)	Gain (μV/ion)	ENC (ions RMS)	Lag-1 correlation
20	19.5441	40.157	50.18	-0.2756
40	19.5440	41.479	58.44	-0.2463

S/N is the noiseless recovered signal divided by frame ENC, evaluated over all positive input pixel–frame cells. The cell-weighted and event-weighted means are shown separately. Compression compares the noiseless transistor response with the calibrated linear prediction; the flags require a deficit of at least one frame ENC. A pad history is counted once per event if any positive-input bucket passes the indicated local-loss criterion.

Sample	Events	ENC	Mean S/N	Event mean S/N	Loss50 events (%)	Deficit (%)	Total bias (%)
Double beta	8000	58.44	5.559	6.423	8.05	0.637	+2.597

The positive-cell compression deficit is $\sum_{N_n>0} \max(\widehat{N}_{\mathrm{linear,n}} - \widehat{N}_{\mathrm{CSA,n}}, 0) / \sum N_n$. The signed total bias is $(\sum \widehat{N}_{\mathrm{CSA,n}} - \sum N_n) / \sum N_n$, including quiet frames and recovery. The linear prediction's own calibration bias is saved separately, as is the noisy summed estimate. The isolated-packet calibration sets the central-frame response to one ion; its earlier and later frame contributions remain in the complete history. Their saved three-tap kernel predicts a total linear charge bias of 3.238%. This identifies the common positive offset in the present scalar estimator. These diagnostics retain that existing estimator and apply no detection threshold.

The fast response is checked against control histories spanning predicted output voltage. Direct SPICE is expanded until remaining control errors fall below the configured 0.5 ENC. The final voltage guard is 0.63 V and the largest remaining control error is 0.3952 ENC. Independent solver refinement changes the sampled response by at most 1.915e-05 ENC. Independent checks cover finite-path charge, spectral integration, covariance convergence and positive definiteness, sampled zero-input mean/variance/covariance, input ion ledgers and Float32 storage errors relative to Float64 calculations.

Response files use `nausicaa.ion_csa_response.pilot.v1`. Each saves the selected IDs, parent file/hash, exact cards, code identity, stable per-pad seed, frame covariance, input counts, mean voltages, linear/noiseless/noisy recovered ions and compression flags. Files are written atomically and marked complete after the requested event ledgers pass.

The companion `ion_csa.html` provides sample and compression-severity selectors, diagnostic distributions and typical/weak/compressed pad histories. S/N is a quality observable; final analysis determines acceptable performance. Spatial noise occupancy requires a finite readout area; the present controls use zero-input histories on the selected channels.

From the repository root:

```
work/event-viewer-venv/bin/python scripts/production/run_ion_csa_pilot.py
# Presentation only:
work/event-viewer-venv/bin/python scripts/production/present_ion_csa_pilot.py \
  --results RESULTS_DIRECTORY
```

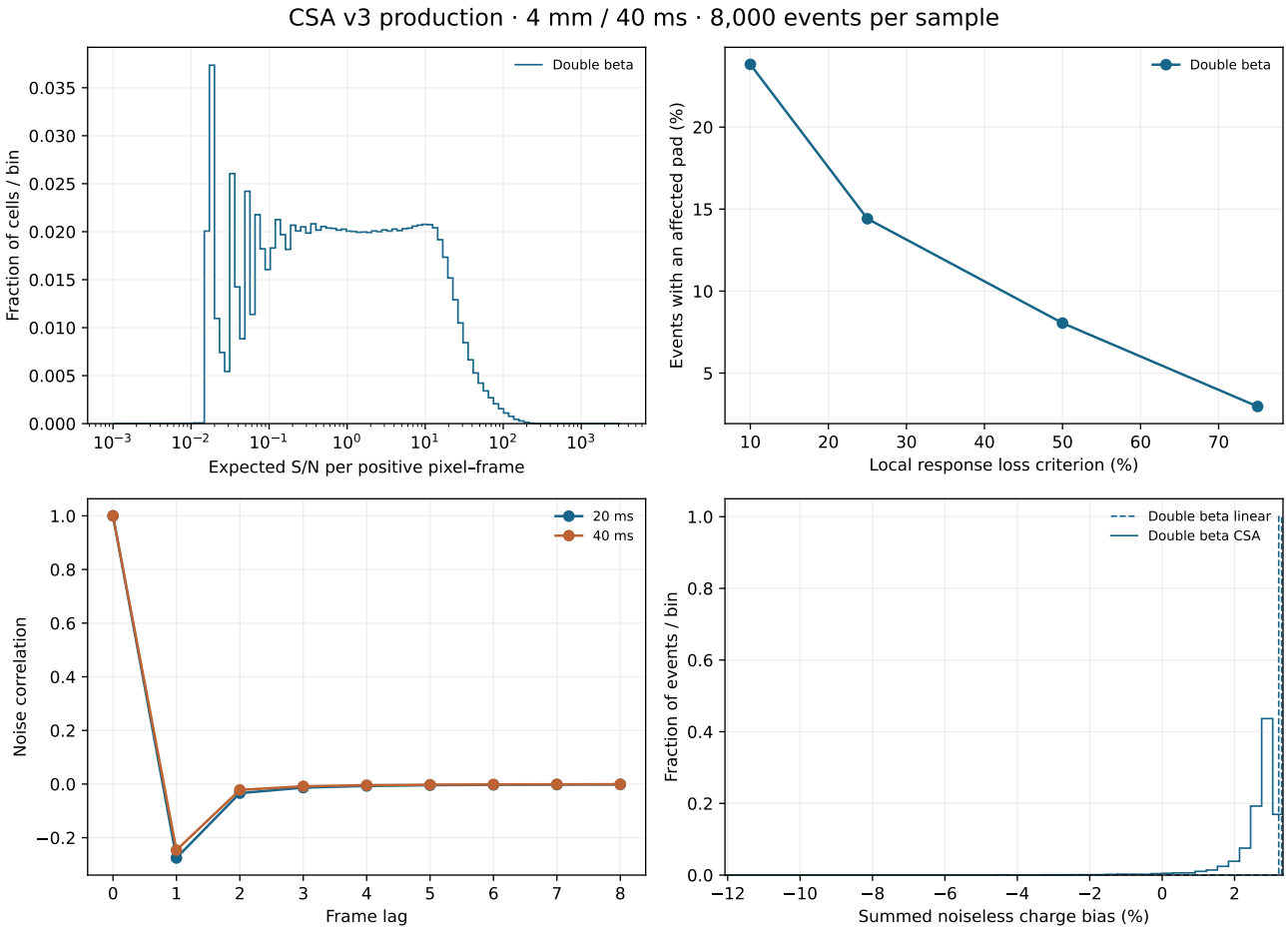


Figure 1: Saved study quality, compression, covariance and charge-bias diagnostics. Dashed charge-bias curves use the linear estimator; solid curves include transistor response where directly simulated.